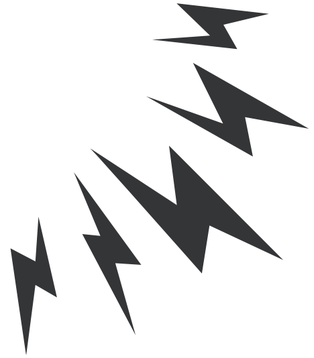
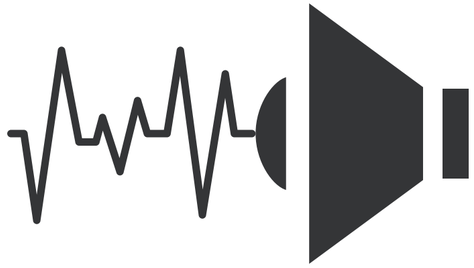


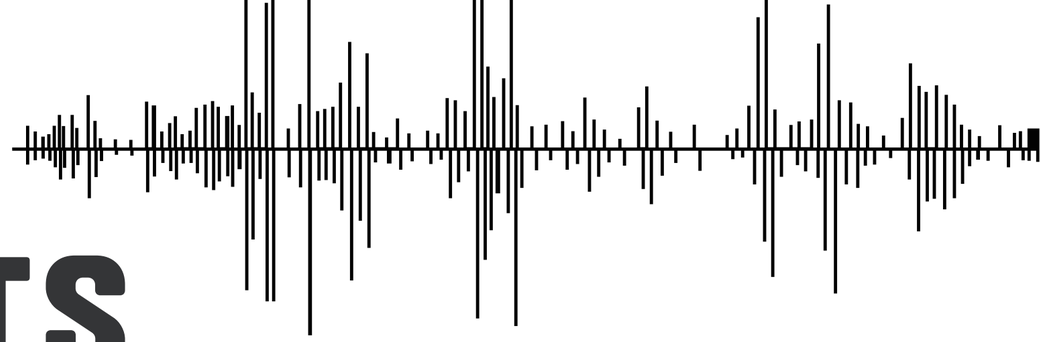


AUDIO NOISE REDUCTION

Digital Signal Processing - Term Project





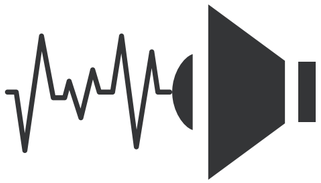


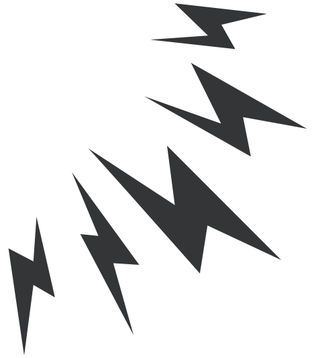
CONTENTS

- Overview
- Audio Preparation
- Exploratory Data Analysis
- Noise reduction pipeline
- Test and validate algorithm with noisy records
- Failure Cases
- Conclusion





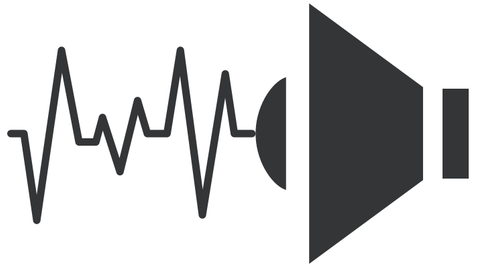






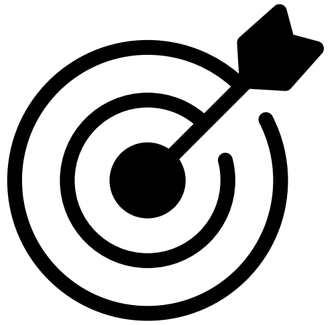
OVERVIEW





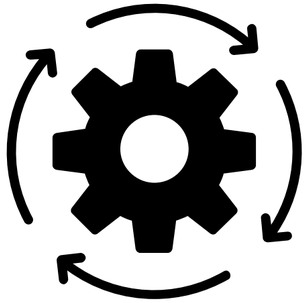
OVERVIEW

1



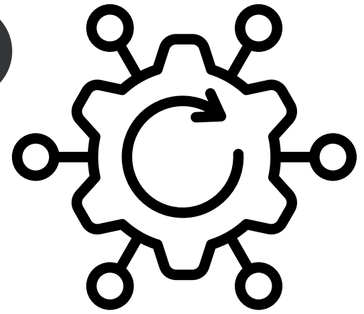
Restore real world noisy audio by DSP algorithm

2



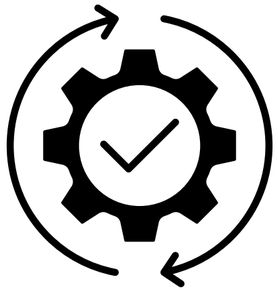
Add custom noise profiles with clean reference audio

3



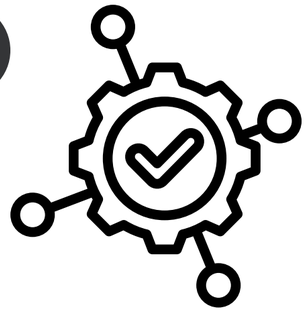
Build DSP pipeline based on the synthetic baseline

4

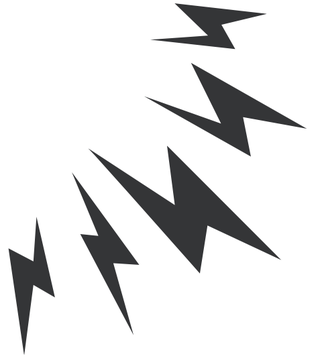


Test the algorithm against unknown noisy records

5



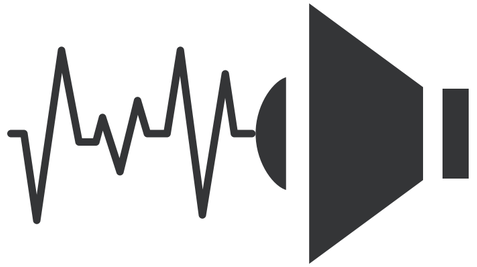
Use algorithm with real world noisy audio



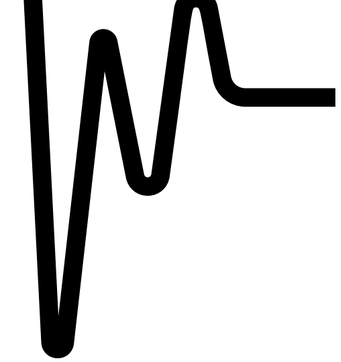


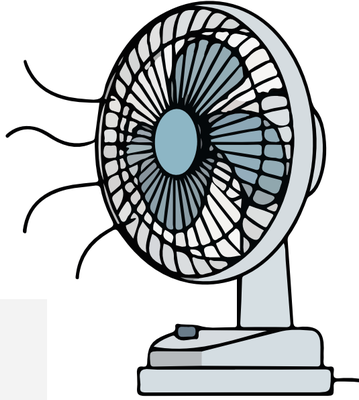
AUDIO PREPARATION





AUDIO PREPARATION



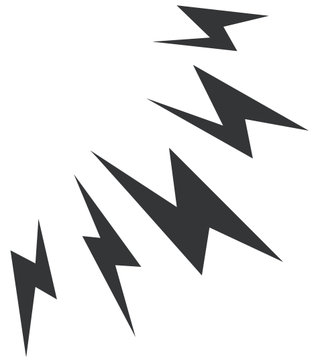


- audio
  - clean_speech.wav
  - hci_audio_record.m4a
  - hci_audio_record.wav
  - noise_recording.m4a
  - noise_recording.wav

- noisy_audio
  - noisy_1.wav
  - noisy_2.wav
  - noisy_3.wav
  - noisy_4.wav
  - noisy_5.wav
  - noisy_6.wav
  - noisy_7.wav
  - noisy_8.wav
  - noisy_9.wav
  - noisy_10.wav



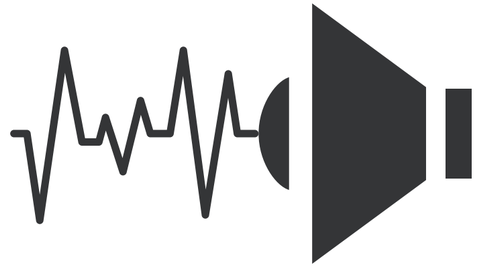






EXPLORATORY DATA ANALYSIS (EDA)





EXPLORATORY DATA ANALYSIS



Convert from .m4a file to .wav file

!ffmpeg -y -loglevel error -i audio/sound_1.m4a -ac 1 -ar 44100 -c:a pcm_s16le audio/sound_1.wav

clean_speech.wav

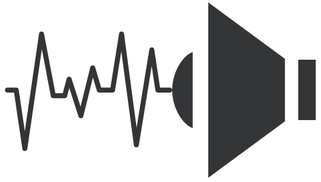
```
fs = 16000
Samples = 172343
Duration = 10.7714375 s
signal shape = (172343,)
signal dtype = int16
```

noise_recording.wav

```
fs = 44100
Samples = 587001
Duration = 13.310680272108844 s
signal shape = (587001,)
signal dtype = int16
```

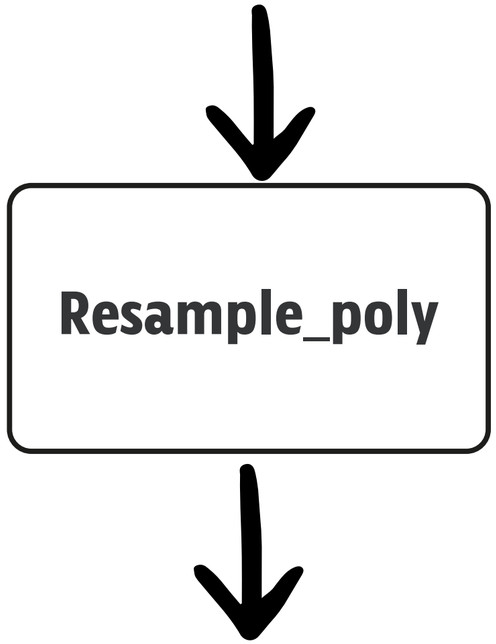


```
1   noise = noise_signal.astype(np.float64)
2   noise = noise / np.max(np.abs(noise))
```



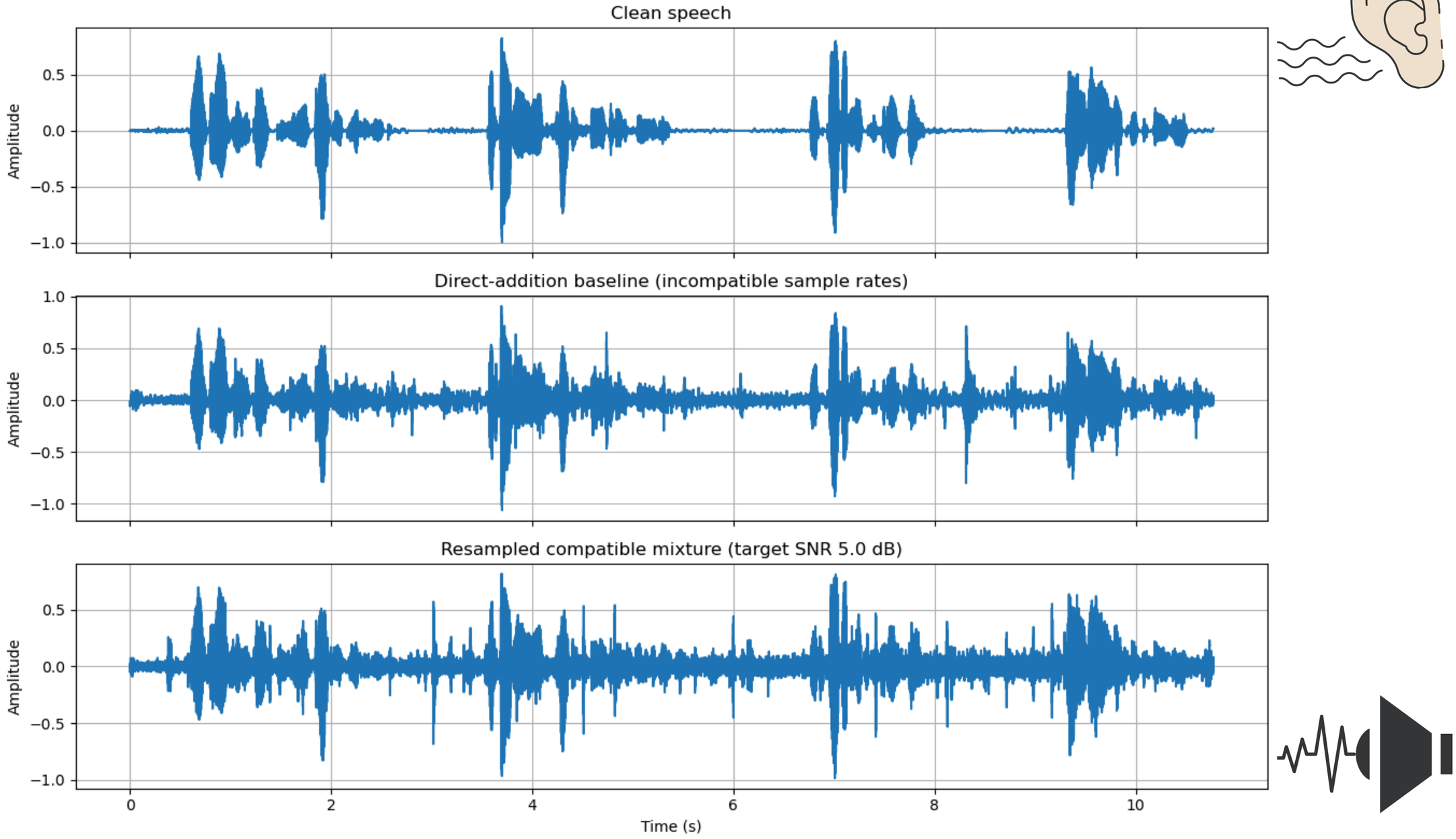
EXPLORATORY DATA ANALYSIS

```
Clean speech: 16000 Hz, 172343 samples (10.77 s)
Noise:        44100 Hz, 587001 samples (13.31 s)
```



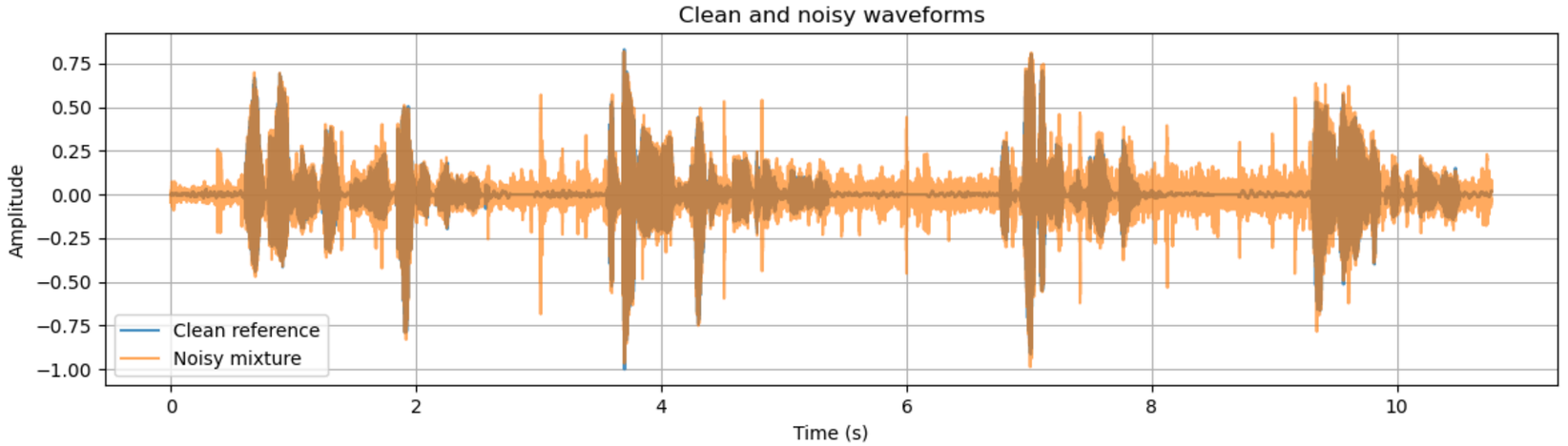

```
Noise sampling rate: 44100 Hz -> 16000 Hz
Baseline samples: 172343
Compatible samples: 172343
Baseline mixture SNR estimate: 5.16 dB
Compatible mixture SNR: 5.00 dB (target 5.00 dB)
```

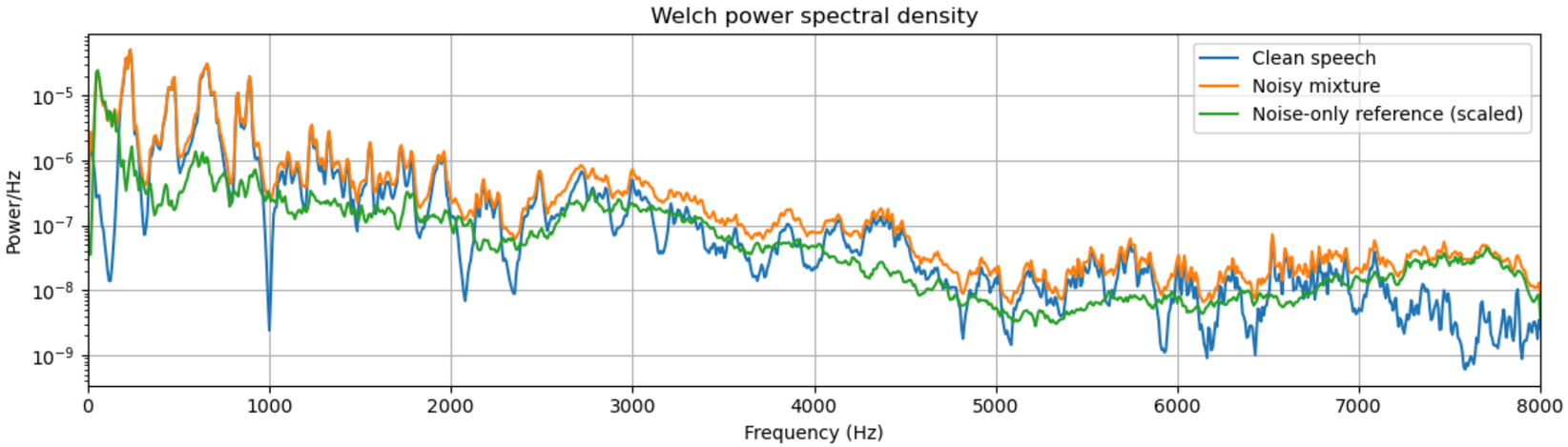
EXPLORATORY DATA ANALYSIS

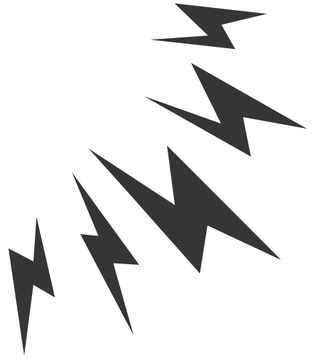


EXPLORATORY DATA ANALYSIS

```
Noise PSD power by candidate filter band:
  0-70 Hz: 0.000510092 (24.32% of estimated noise power)
  70-7500 Hz: 0.00145155 (69.21% of estimated noise power)
  7500-8000 Hz: 1.21106e-05 (0.58% of estimated noise power)
```



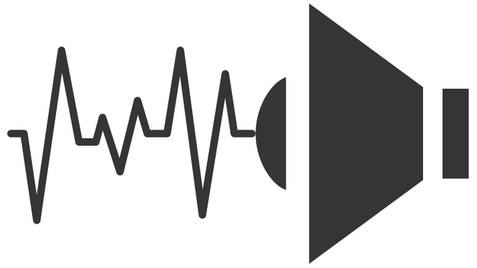






NOISE REDUCTION PIPELINE



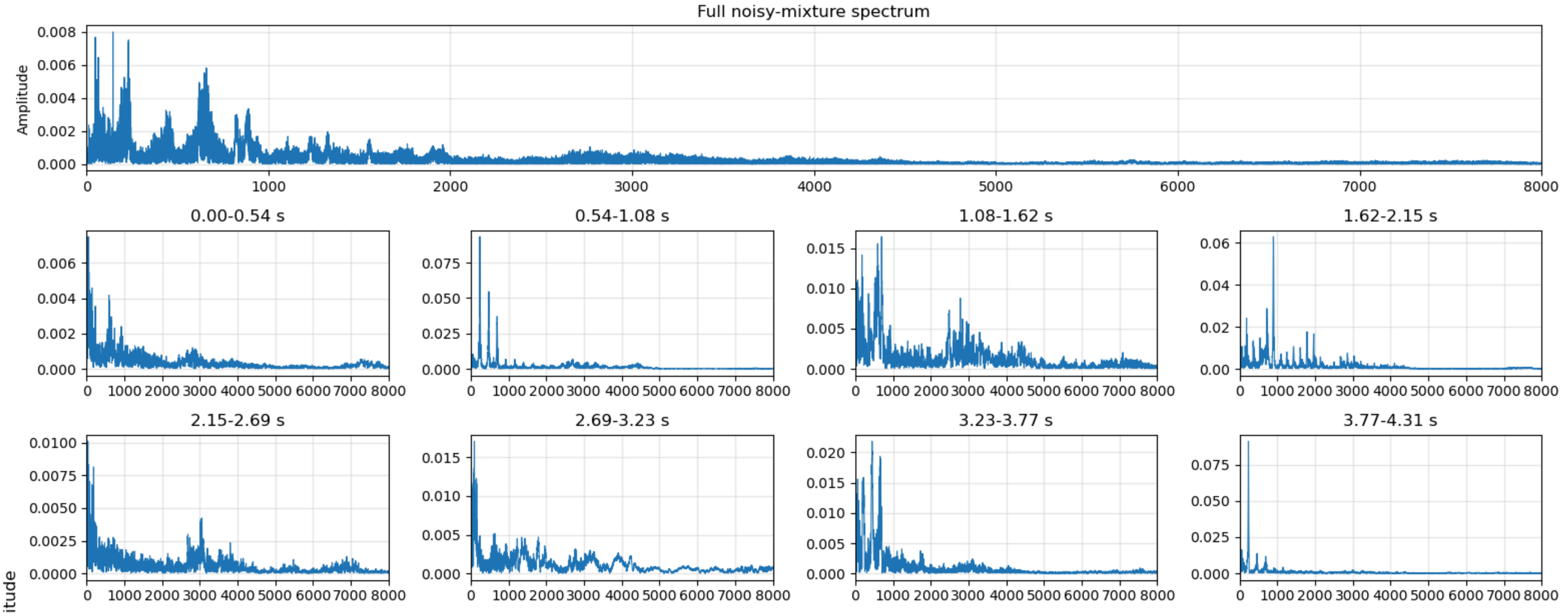


FFT SPECTRUM





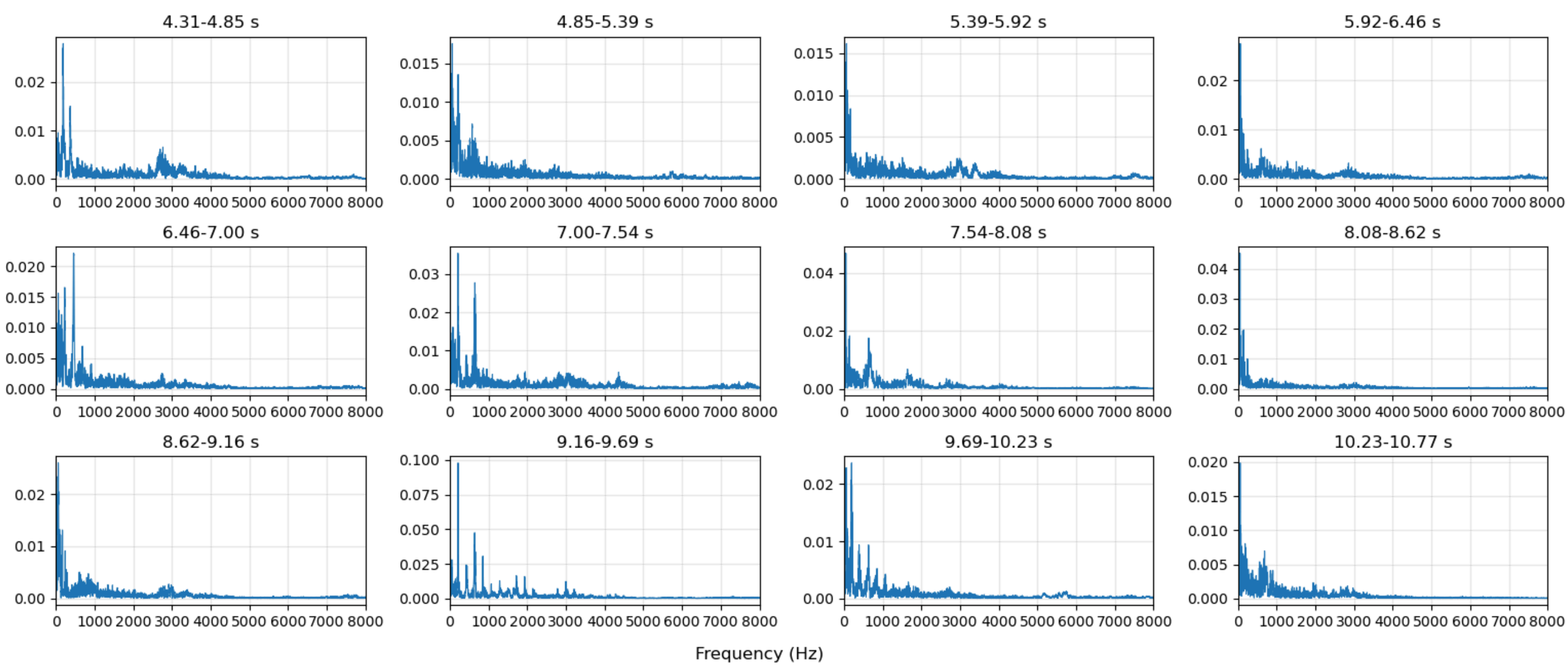
Noisy mixture full spectrum and 20 time frames



FFT SPECTRUM



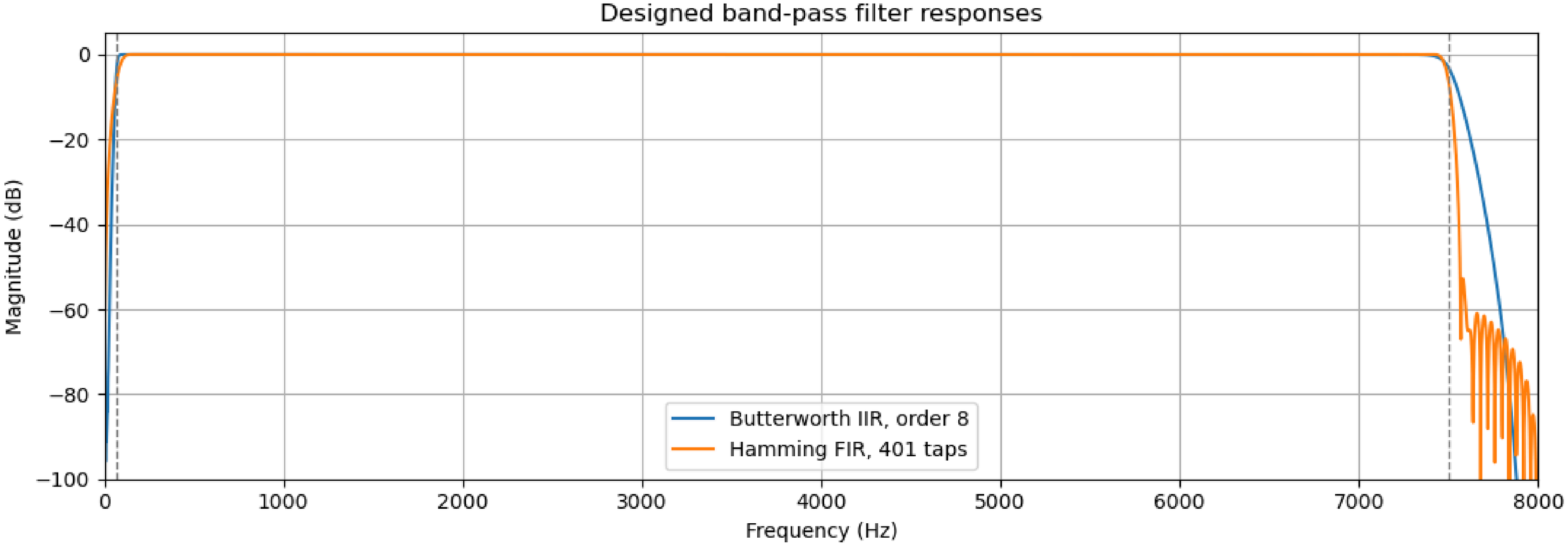




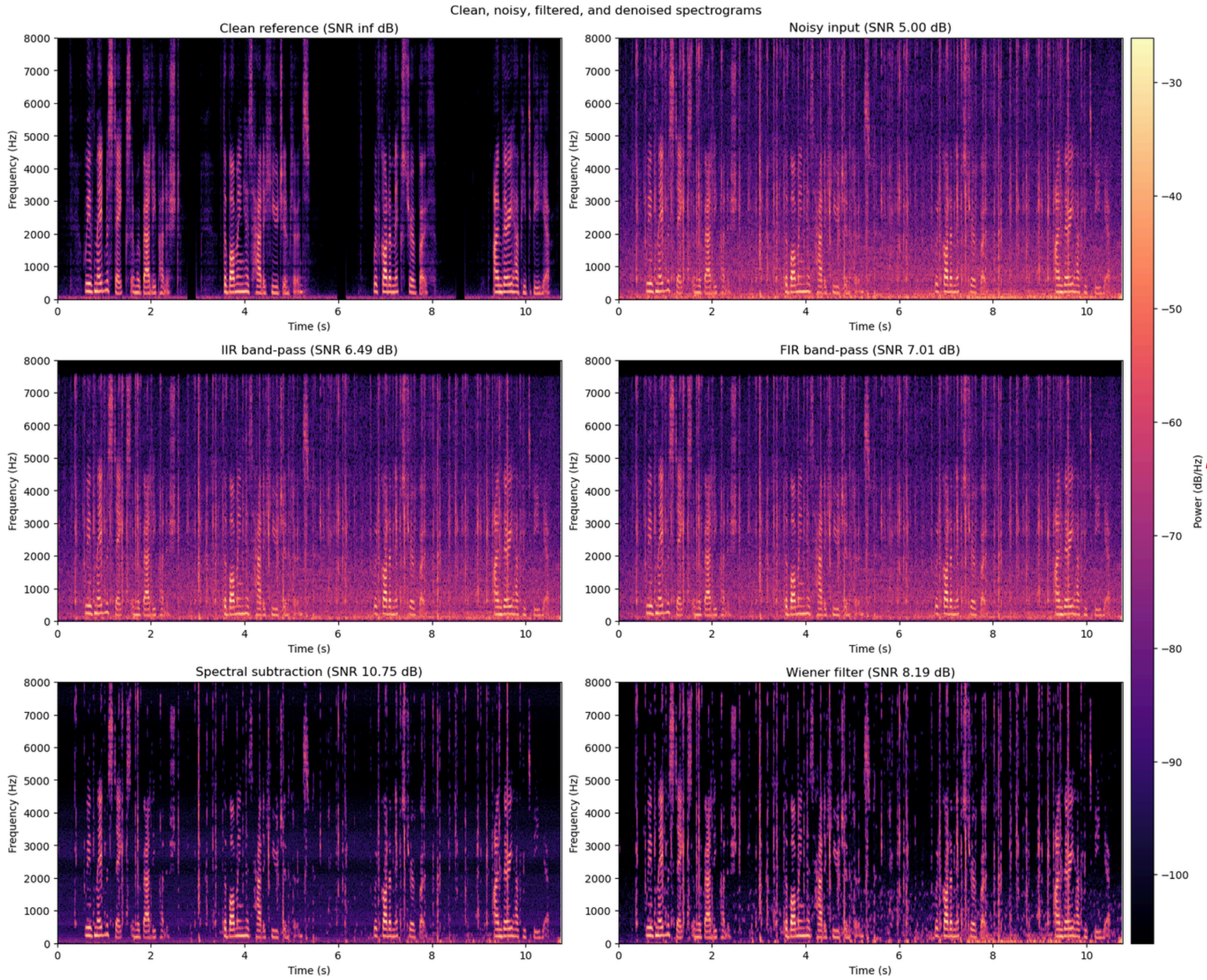
FIR AND IIR BASELINE FILTER







STFT

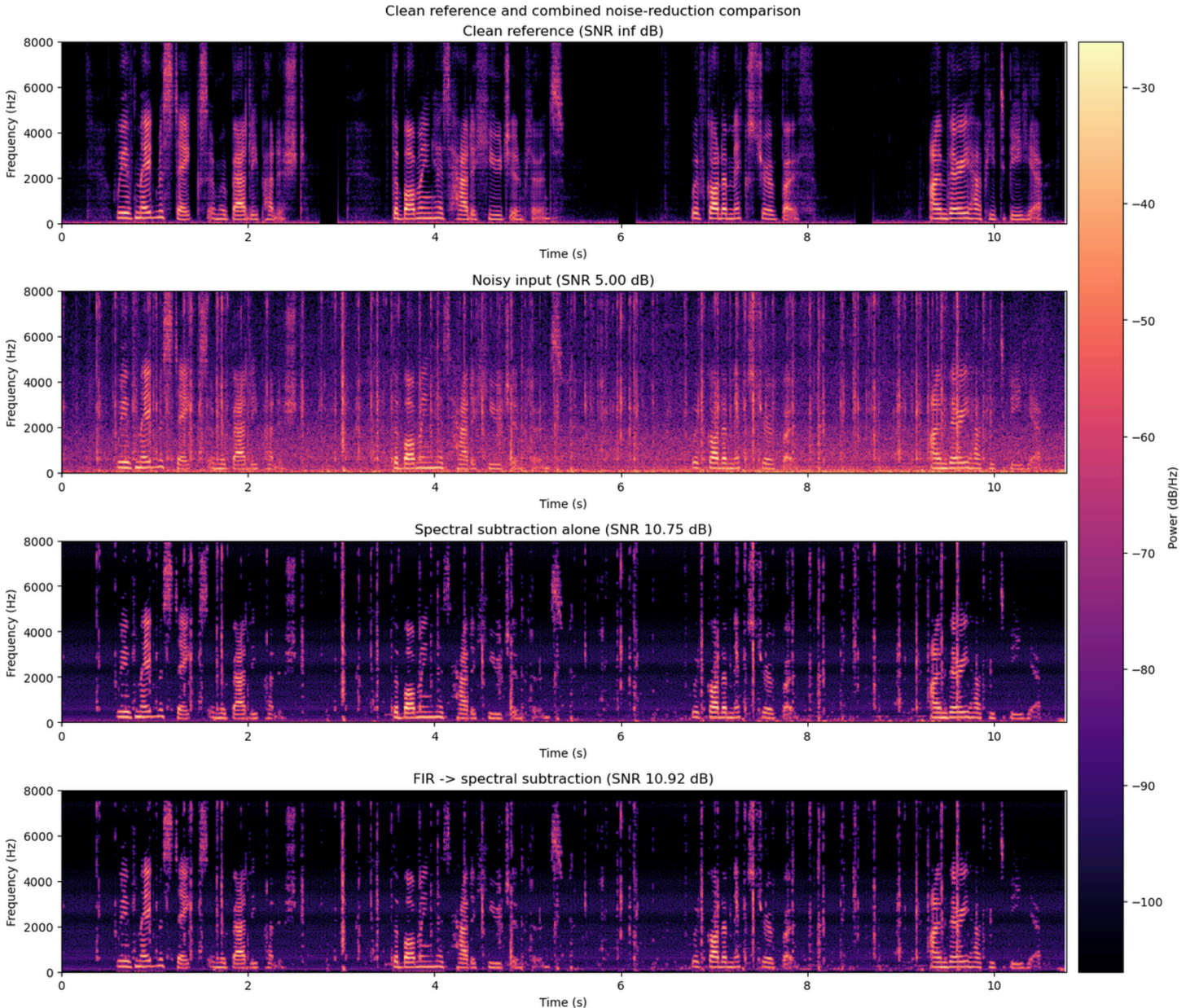




Measured input SNR: 5.00 dB

Method	Output SNR	Improvement
Noisy input	5.00 dB	+0.00 dB
IIR band-pass	6.49 dB	+1.49 dB
FIR band-pass	7.01 dB	+2.01 dB
Spectral subtraction (estimated profile)	10.75 dB	+5.75 dB
Wiener filter	8.19 dB	+3.19 dB
Oracle spectral subtraction	14.45 dB	+9.45 dB

COMBINED DESOISING PIPELINE





```
Standalone FIR SNR: 7.01 dB
Standalone spectral subtraction SNR: 10.75 dB
Combined pipeline                                  Output SNR    Improvement vs input
-------------------------------------------------------------------------------------
Noisy input                                          5.00 dB          +0.00 dB
FIR -> spectral subtraction                         10.92 dB          +5.92 dB
IIR -> spectral subtraction                         10.83 dB          +5.83 dB
Oracle FIR -> spectral subtraction                  13.96 dB          +8.96 dB
FIR -> spectral subtraction change vs standalone spectral subtraction: +0.17 dB
```

TARGETED KEYBOARD CLICK EXPERIMENT



1. Find click:
   - Use the known noise track
   - Band-pass the audio
   - Use median over frame
   - Clean up the detections
2. Spectral subtraction
   - STFT
   - Noise profile
   - Subtract

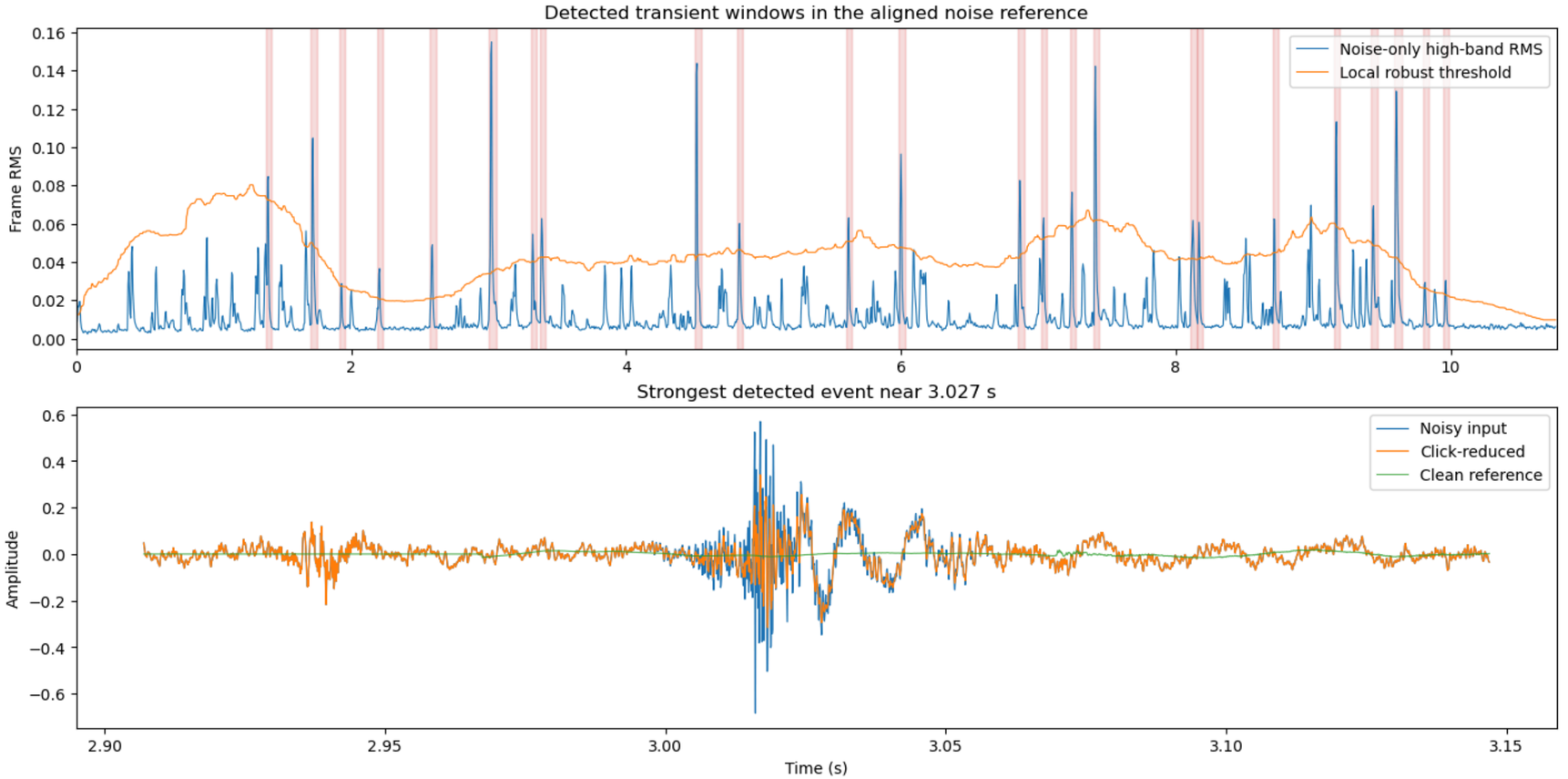
Goal:

- removes clicks that would damage the spectral subtraction stage
- needs no clean signal or noise recording
- tells what kind of noise each file has

TARGETED KEYBOARD CLICK EXPERIMENT

```
Detected transient events: 24
Total event-window coverage: 9.9% of the recording
Click attenuation target in 1500-7500 Hz: -12.0 dB
```

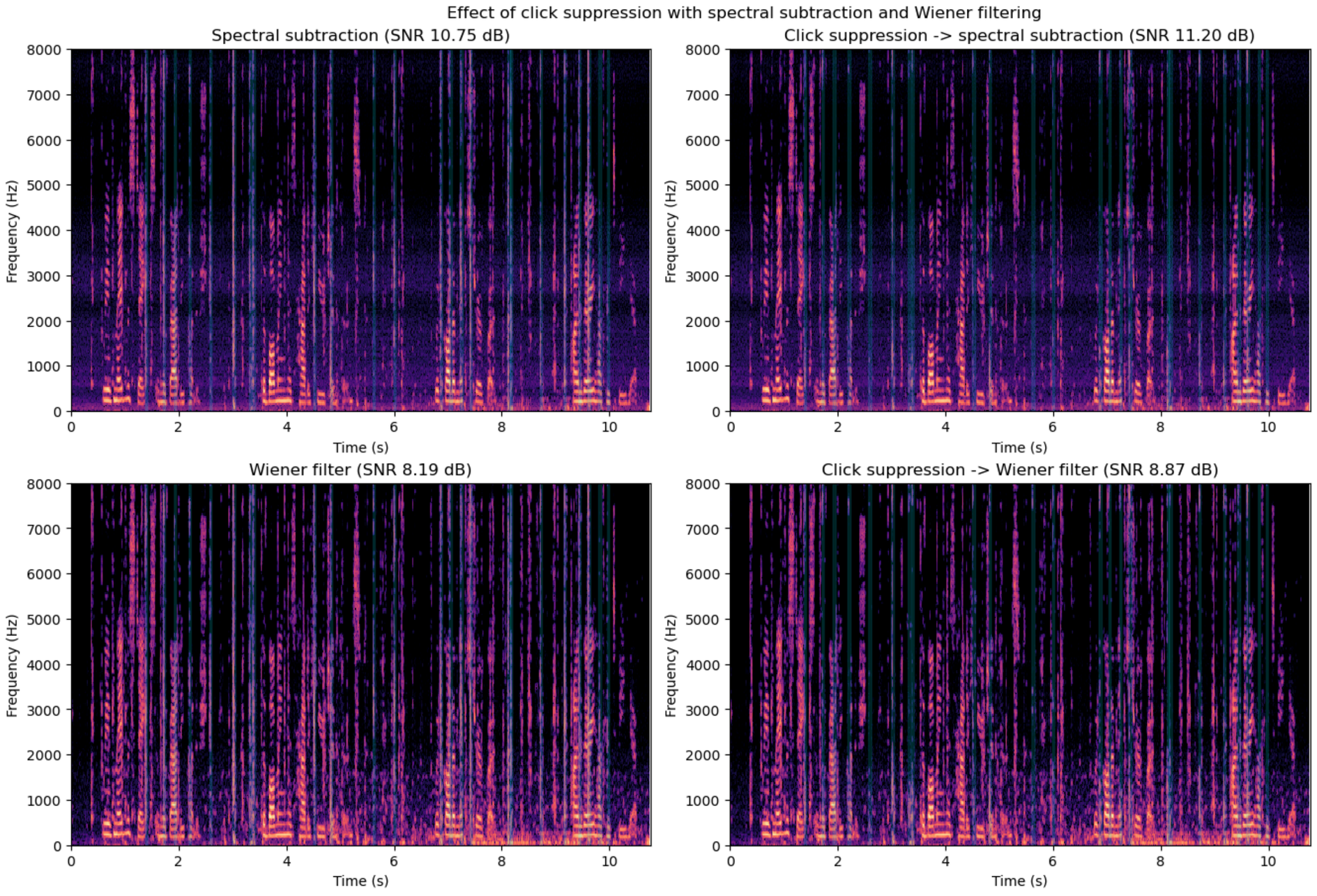
Method	Overall SNR	Event-band reduction
Noisy input	5.00 dB	0.00 dB
Localized click suppression	5.37 dB	6.62 dB
Spectral subtraction	10.75 dB	3.19 dB
Click suppression -> spectral subtraction	11.20 dB	6.86 dB
Wiener filter	8.19 dB	0.66 dB
Click suppression -> Wiener filter	8.87 dB	7.22 dB

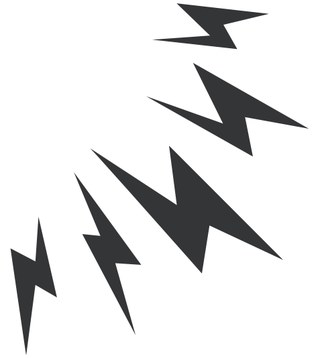


TARGETED KEYBOARD CLICK EXPERIMENT





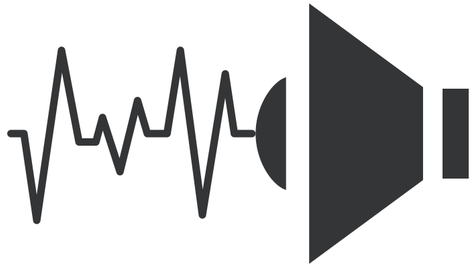






TEST AND VALIDATE ALGORITHM WITH NOISY RECORDS

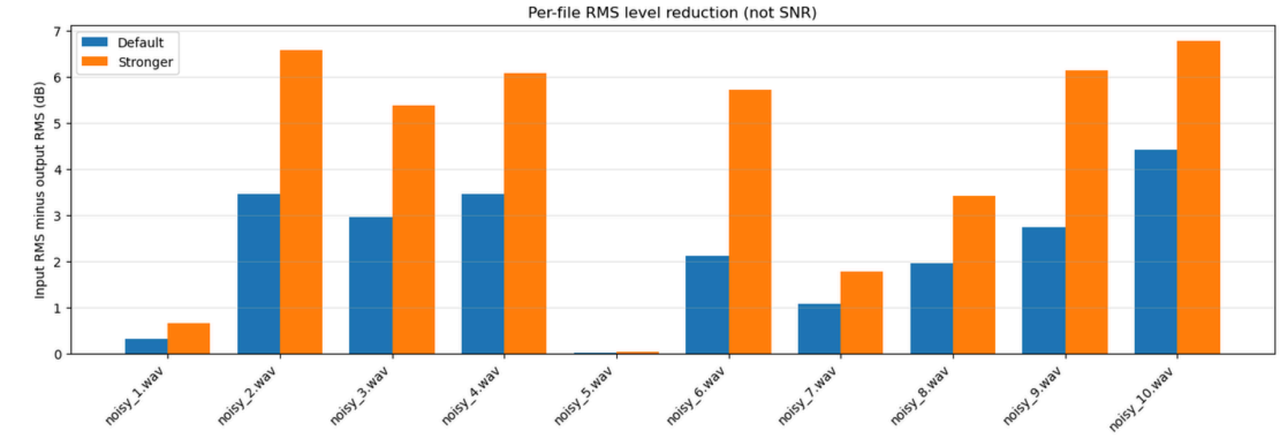


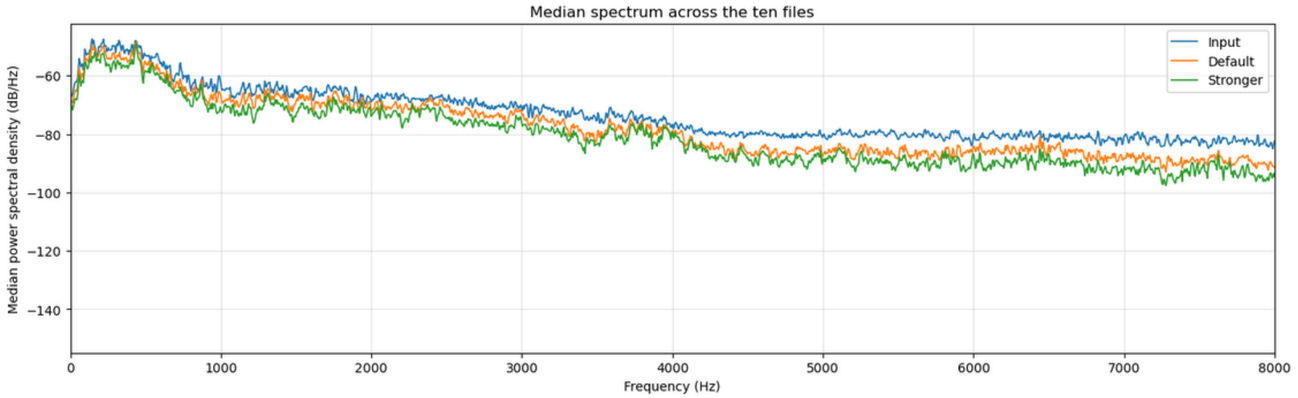


VALIDATE ALGORITHM WITH 10 NOISY RECORDS

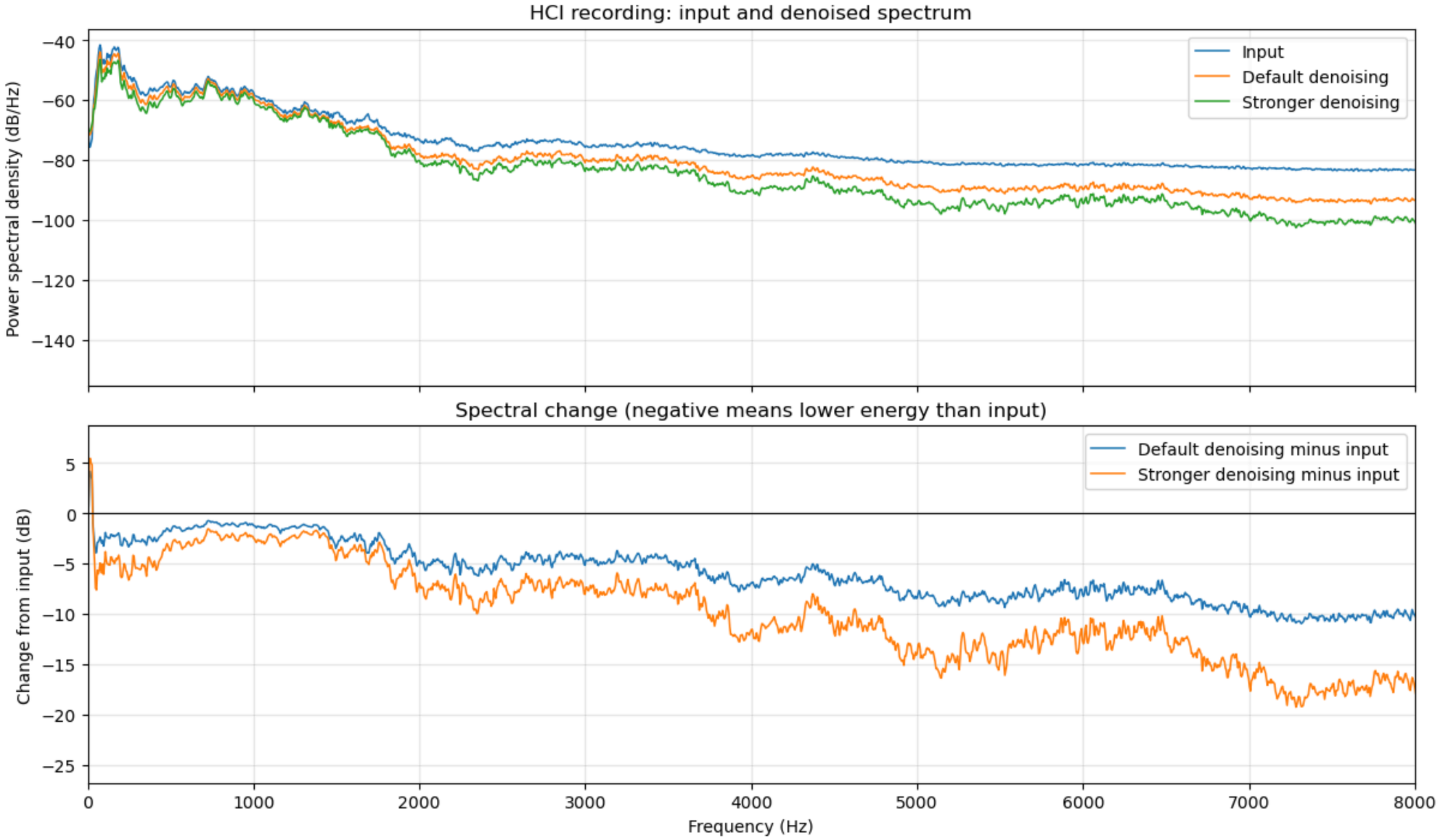
RMS changes are level differences, not SNR improvements.

File	fs (Hz)	Duration	Input RMS	Default RMS	Change	Strong RMS	Change	Click candidates
noisy_1.wav	22050	10.00s	-22.93 dBFS	-23.24 dBFS	-0.31 dB	-23.60 dBFS	-0.67 dB	2/2
noisy_2.wav	22050	10.00s	-17.13 dBFS	-20.59 dBFS	-3.46 dB	-23.71 dBFS	-6.58 dB	0/0
noisy_3.wav	22050	10.00s	-13.19 dBFS	-16.14 dBFS	-2.95 dB	-18.57 dBFS	-5.38 dB	1/1
noisy_4.wav	22050	10.00s	-15.60 dBFS	-19.05 dBFS	-3.45 dB	-21.66 dBFS	-6.07 dB	0/0
noisy_5.wav	22050	10.00s	-14.23 dBFS	-14.26 dBFS	-0.03 dB	-14.27 dBFS	-0.04 dB	31/31
noisy_6.wav	22050	10.00s	-20.24 dBFS	-22.35 dBFS	-2.12 dB	-25.95 dBFS	-5.71 dB	8/8
noisy_7.wav	22050	9.47s	-21.99 dBFS	-23.07 dBFS	-1.08 dB	-23.77 dBFS	-1.78 dB	2/2
noisy_8.wav	22050	10.00s	-22.81 dBFS	-24.77 dBFS	-1.95 dB	-26.24 dBFS	-3.42 dB	5/5
noisy_9.wav	22050	10.00s	-16.74 dBFS	-19.49 dBFS	-2.75 dB	-22.87 dBFS	-6.13 dB	19/19
noisy_10.wav	22050	10.00s	-19.45 dBFS	-23.87 dBFS	-4.42 dB	-26.23 dBFS	-6.78 dB	1/1

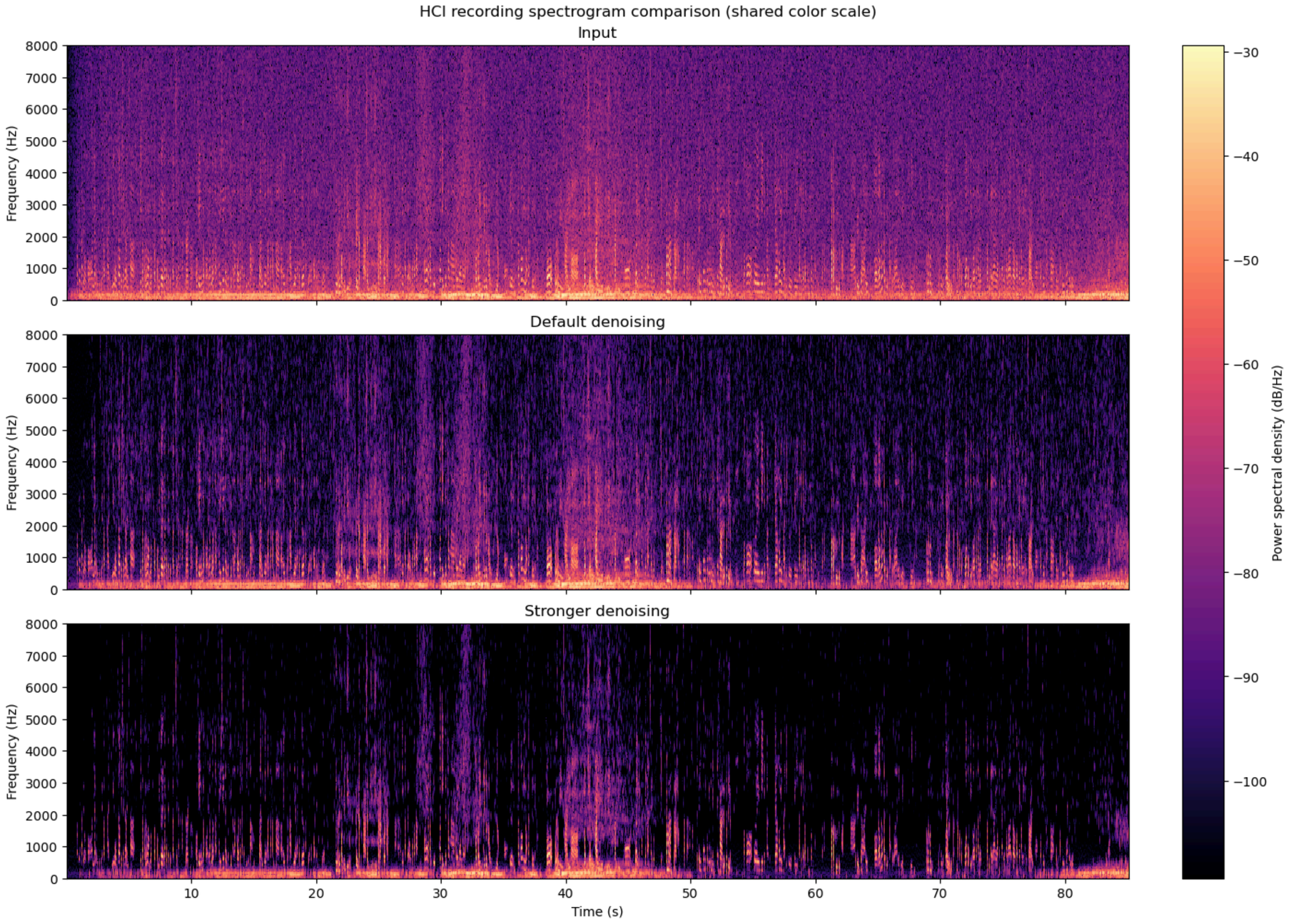


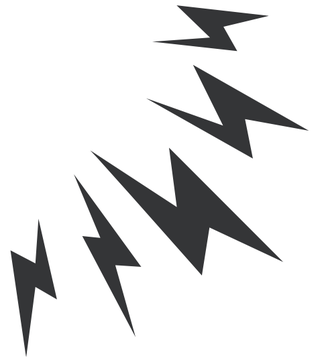


TEST ALGORITHM WITH 10 NOISY RECORDS



TEST ALGORITHM WITH 10 NOISY RECORDS

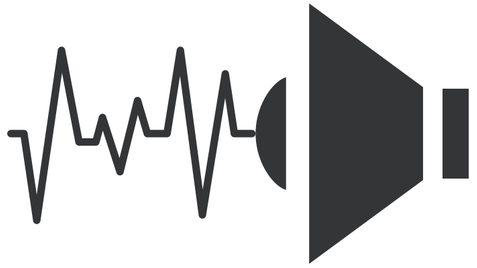






FAILURE CASE

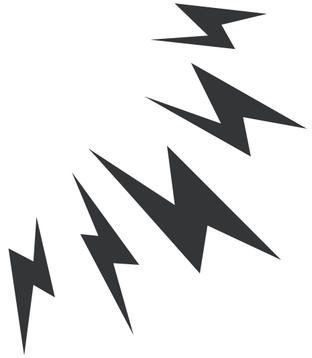




FAILURE CASES



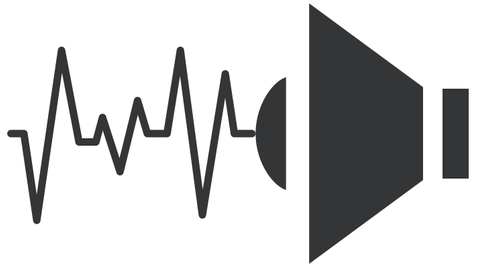
- **FIR/IIR filtering:** Limited when noise overlaps speech frequencies
- **Click detection:**May mistake speech transients for clicks and suppress wanted speech.
- **Aggressive denoising:** Can reduce speech quality along with noise.
- **Evaluation limitation:** No clean reference for the unknown and batch recordings, so noise-removal quality cannot be objectively confirmed.





CONCLUSION

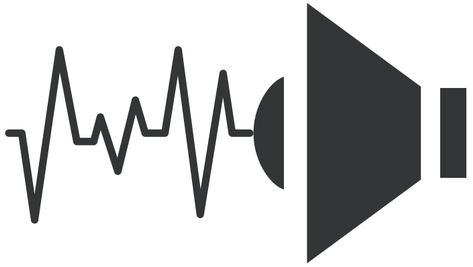




CONCLUSION

- Spectral subtraction performed best in controlled testing (+5.75 dB SNR).
- Filter FIR/IIR can’t use in this noise reduction project
- Default click suppression plus spectral subtraction sounded best on the real recording, but is not yet effective enough for real world noise.
- Further improvement and testing are needed.







MEMBER

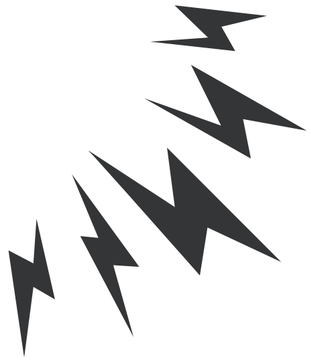
670710123 กัญญาณัฐ เขมนรากร
670710258 Sothea Sokea

REFERENCE

- clean speech: https://www.kaggle.com/datasets/anhquocdevelop/clean-speech
- noisy audio: https://www.kaggle.com/datasets/367a72ddd73170fa1dee00402d00fcb5e0cbe08d84c0db192a69176bdf8606ee
- scipy : https://docs.scipy.org/doc/scipy/index.html
- Digital Signal Processing course lessons

AI USAGE

- **Github Copilot:** helped edit and debug notebook code, explain DSP concepts, and document the project
- **Claude:** used to verify approaches and explain concepts or results.
- **ChatGPT:** used to check code and explanations.

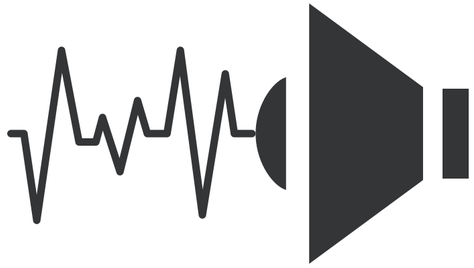


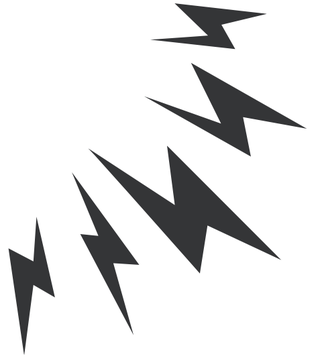
MORE INFOMATION



github repository: audio_noise_reduction









THANK YOU



